

KLK2

kallikrein related peptidase 2 *Synonyms: KLK2A2, hGK-1, hK2*

HGNC HGNC:6363 UNIPROT P20151 NCBI GENE 3817 ENSEMBL ENSG00000167751

SOURCES · 0/5 UniProt GO HPA SURFY CSPA

BENCHMARK CONTEXTUAL agrees with deep dive

TRIAGE Contextual — initial pass · NO WEB SEARCH · agrees with deep dive

KLK2 (human kallikrein-2) is a canonical secreted serine protease with no transmembrane domain, but a 2025 preclinical study (PMC12580770) demonstrates surface accessibility on intact prostate cancer cells by live-cell FACS on VCaP and fresh mCRPC patient tumor cells, confocal IF, and functional engagement via three distinct therapeutic modalities. Surface expression is strictly prostate-lineage- and AR-state-restricted, absent in neuroendocrine/double-negative mCRPC variants. The primary risk is a large competing secreted pool (KLK2 is constitutively secreted into conditioned medium and seminal plasma), and the surface-docking mechanism remains uncharacterized.

SURFACE VERDICT

Moderate

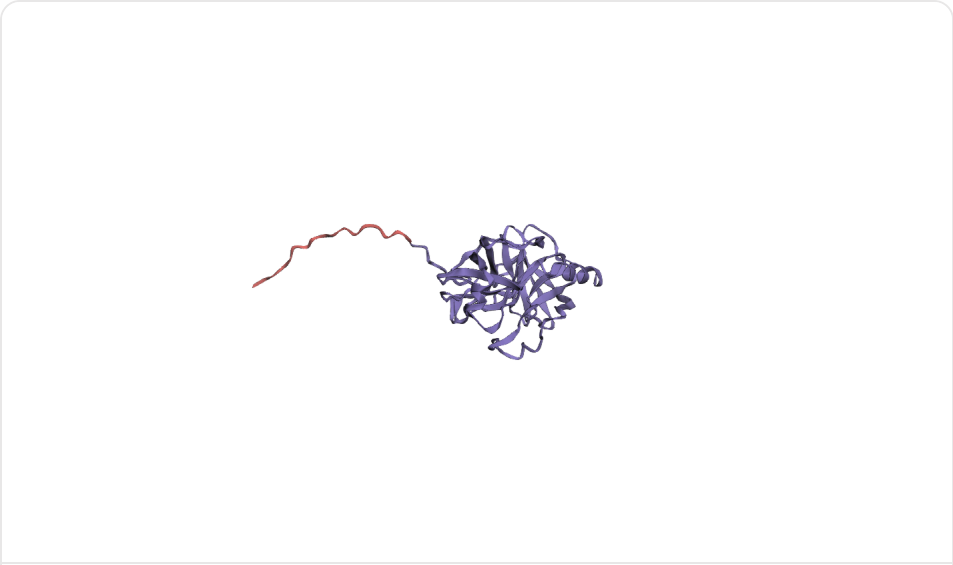
Architecture · Other

CONFIDENCE

Moderate

22 primary · 11 secondary

✓ KNOWN LIGAND



AlphaFold AFDB P20151 · v6

Extracellular · Signal peptide

SONNET · CONTEXTUAL · TISSUE RESTRICTED SURFACE

SRC

SRC proto-oncogene, non-receptor tyrosine kinase *Synonyms: ASV, SRC1, THC6*

HGNC HGNC:11283 UNIPROT P12931 NCBI GENE 6714 ENSEMBL ENSG00000197122

SOURCES · 2/5 UniProt GO HPA SURFY CSPA

BENCHMARK CONTEXTUAL agrees with deep dive

TRIAGE Contextual — initial pass · NO WEB SEARCH · agrees with deep dive

SRC is state-dependently surface-accessible in cancer cells — a canonical inner-leaflet kinase with a cancer-specific outer-surface form. Direct surface evidence is single-method: antibody-mediated tumor cell killing against extracellular-facing eSrc in cancer cell lines and xenograft models demonstrates outer-leaflet engagement. Surface presence is strictly state-gated, requiring cancer-state autophagolysosomal exocytosis (ALE) to invert the N-myristoylated kinase onto the outer leaflet; normal cells retain exclusively inner-leaflet, cytoplasmic-face localization.

SURFACE VERDICT

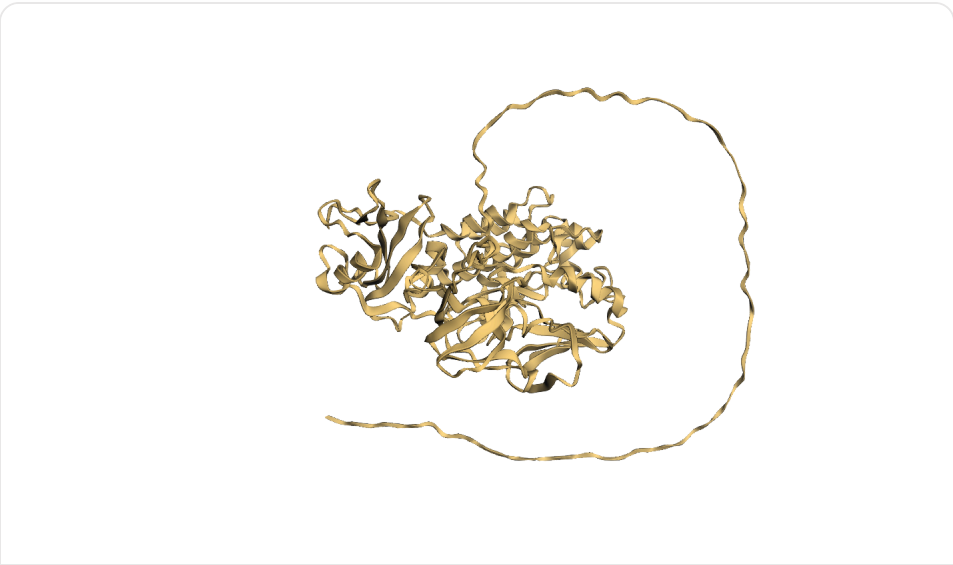
Moderate

Architecture · Other

CONFIDENCE

Low

✓ TUMOR ASSOCIATED



AlphaFold AFDB P12931 · v6

Globular

SONNET · NO · INNER LEAFLET ANCHORED

CD63

CD63 antigen *Synonyms: AD1, HOP-26, ME491*

HGNC HGNC:1692 UNIPROT P08962 NCBI GENE 967 ENSEMBL ENSG00000135404

SOURCES · 4/5 UniProt GO HPA SURFY CSPA

BENCHMARK CONTEXTUAL agrees with deep dive

TRIAGE Contextual — initial pass · NO WEB SEARCH · agrees with deep dive

CD63 is state-dependently surface-accessible as a pan-tissue tetraspanin whose dominant steady-state localization is lysosomal/endosomal but whose surface pool expands markedly upon cellular activation. Direct multi-method support: KO-validated live-cell flow cytometry on HEK293T cells and primary human granulocytes — basophils and eosinophils — across multiple independent studies. Surface levels are highly state-modulated: absent-to-low on resting granulocytes and stromal cells, rising sharply after IgE-mediated degranulation, platelet activation, TCR stimulation, and in disease states including HES, NPC, and the CRC tumor microenvironment.

SURFACE VERDICT

High

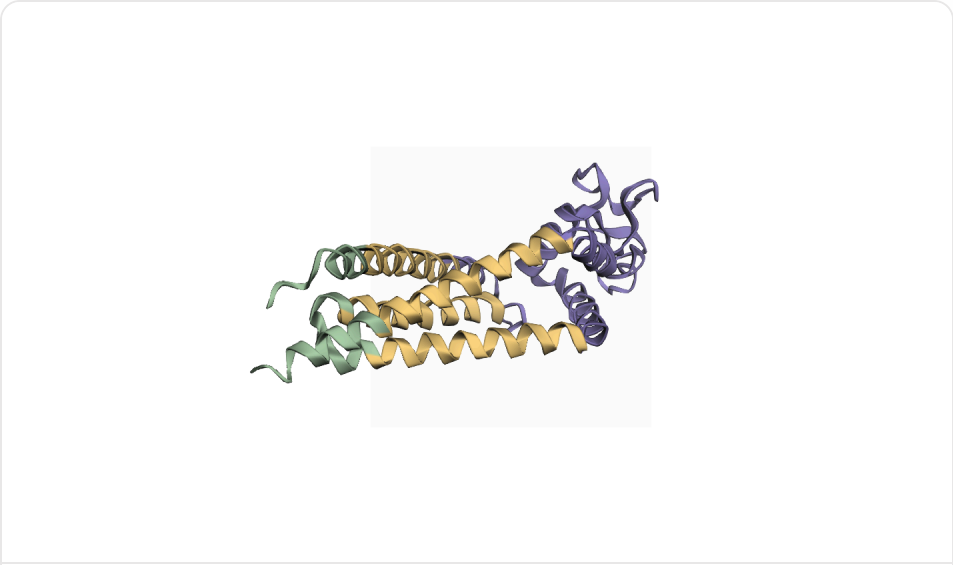
Architecture · Tetraspanin

CONFIDENCE

High

✓ KNOWN LIGAND

PRIMARY · LYSOSOME



AlphaFold AFDB P08962 · v6

Extracellular · TM helix · Intracellular

SONNET · CONTEXTUAL · LYSOSOMAL EXOCYTOSIS